

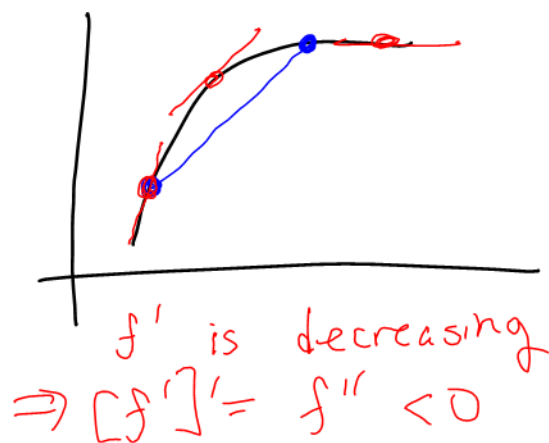
Lecture 18: Concavity and 2nd Derivative Test

Goal: Use the 2nd derivative to determine properties of the original function.

Concave Up [CU]



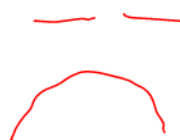
Concave Down [CD]



Theorem (C Test) Let f be a function where f'' exists on an open interval I

① If $f''(x) > 0$ for $x \in I$, f is CU on I .

② If $f''(x) < 0$ for $x \in I$, f is CD on I .



Example 1 Discuss the shape of $f(x) = x^3 - 12x + 1$

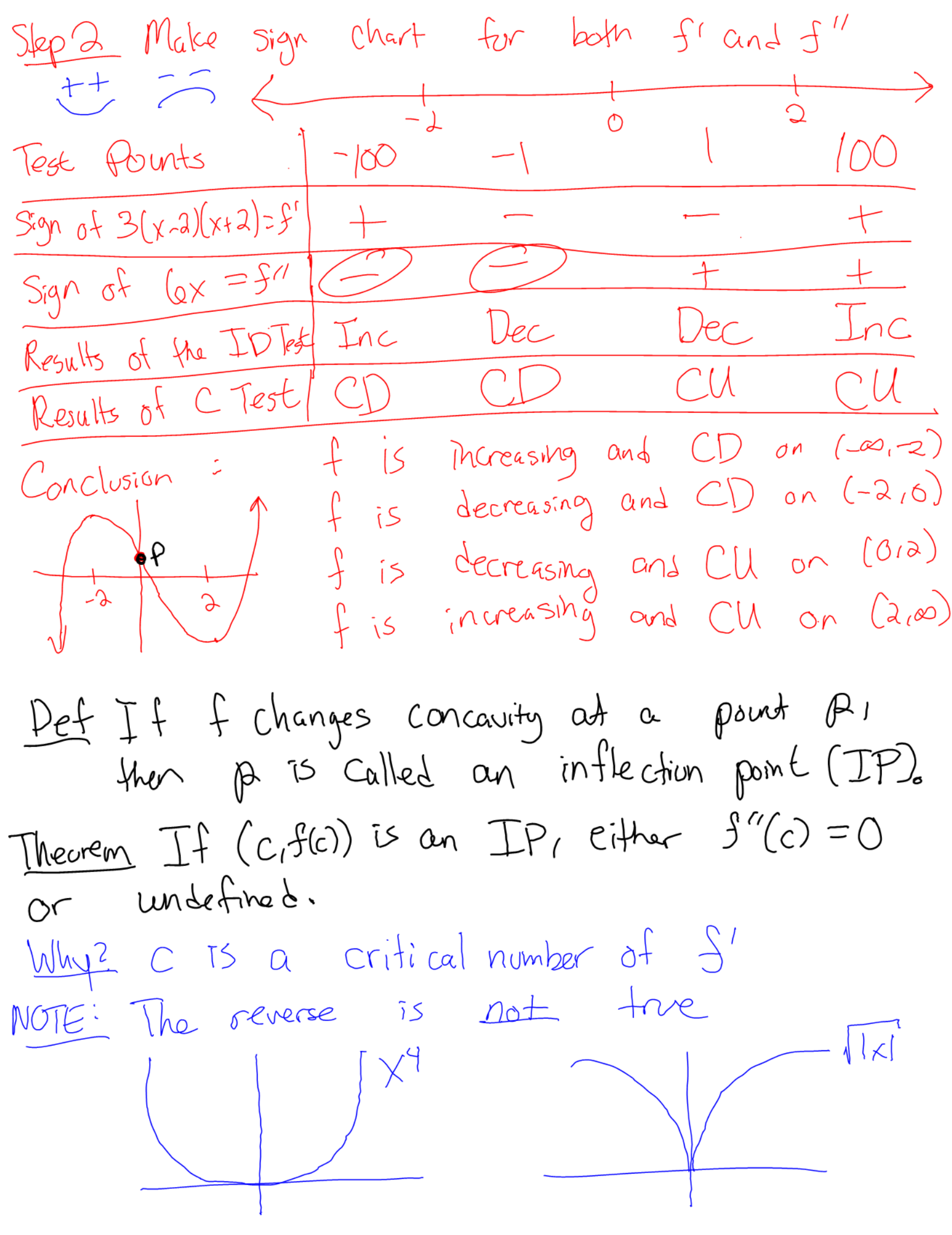


Step 1 Find when $f' = 0$ and $f'' = 0$.
 $f'(x) = 3x^2 - 12 = 3(x-2)(x+2) \underline{\underline{\text{set } 0}}$

$$\Rightarrow x = \pm 2$$

$$f''(x) = 6x \underline{\underline{\text{set } 0}}$$

$$\Rightarrow x = 0$$



Example 2 Determine the locations of all IP for $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$

Step 1 Find when f'' is 0 or undefined

$$f'(x) = 12x^3 - 12x^2 + 12x + 12$$

$$f''(x) = 12 \cdot 3x^2 - 12 \cdot 2x + 12 \cdot 1$$

$$= 12(3x^2 - 2x + 1)$$

$$= 12(x-1)(3x+1) \stackrel{\text{set}}{=} 0$$

$\Rightarrow x = -\frac{1}{3}, 1$ are the only possible locations.

Step 2 Make sign chart for f''

					
Test Points	-100	$-\frac{1}{3}$	0	1	100
Sign of 12	+		+		+
Sign of $x-1$	-		-		+
Sign of $3x+1$	-		+		+
Sign of f''	+		-		+
Results of CTest	CU		CD		CU

Conclusion: f has IPs at $x = -\frac{1}{3}$ and $x = 1$

Example 3 Find all IPs of

$$f(x) = \frac{1}{20}x^5 - \frac{1}{6}x^4 + \frac{1}{6}x^3 - x + 42$$

Step 1

$$f'(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 - 1$$

$$f''(x) = x^3 - 2x^2 + x$$

$$= x(x^2 - 2x + 1)$$

$$= x(x-1)^2 \stackrel{\text{set}}{=} 0$$

$$\Rightarrow x = 0, 1$$

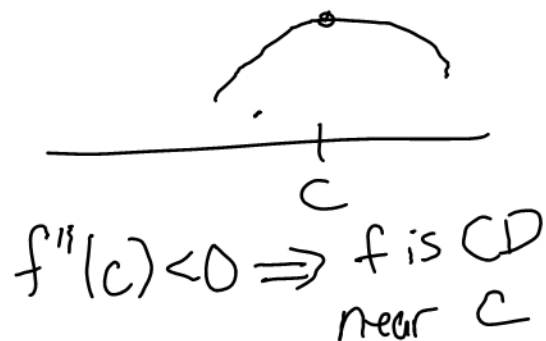
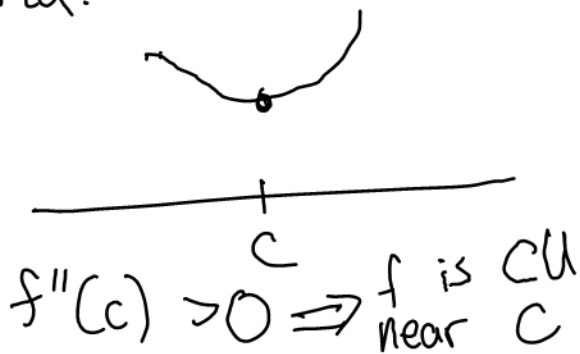
Step 2



Test Point	-100	0	$\frac{1}{2}$	1	100
Sign of x	-		+		+
Sign of $(x-1)^2$	+		+		+
Sign of f''	-		+		+
Results of C Test	CD		CU		CU

$\Rightarrow f$ has only one IP at $(0, f(0)) = (0, 42)$

Concavity can be used to detect relative
extrema. Let c be a critical number of
 f



Theorem (2nd Derivative Test) Let f'' be continuous at c and $f'(c) = 0$.

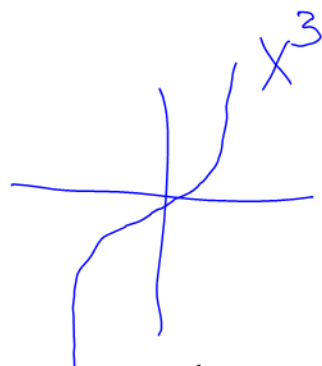
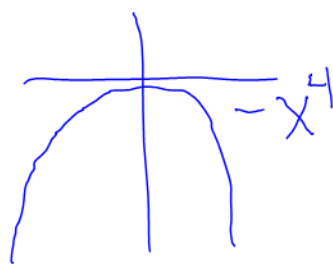
① If $f''(c) > 0$, then f has a rel. min. at c

② If $f''(c) < 0$, then f has a rel. max at c

③ If $f''(c) = 0$, the test is inconclusive.

[I.e., pick another test]

Why ③?



Example 4 locate and classify all relative extrema for $f(x) = x^3 - 3x^2$

Step 1 Solve $f'(x) = 0$

$$f'(x) = 3x^2 - 6x = 3x(x-2) \stackrel{\text{set}}{=} 0$$

$$\Rightarrow x = 0, 2$$

Step 2 Determine the sign of $f''(0)$ and $f''(2)$

$$f''(x) = 6x - 6$$

$$f''(0) = -6 < 0$$

$$f''(2) = 6 > 0$$



Conclusion: f has a relative maximum at $x=0$ and a relative minimum at $x=2$.

Example 5 locate and classify all relative extrema for $f(x) = e^x(x^2 - 5x - 13)$

① Find Crit. Nums.

$$\begin{aligned} f'(x) &= e^x(x^2 - 5x - 13) + e^x(2x - 5) \\ &= e^x(x^2 - 3x - 18) \\ &= e^x(x - 6)(x + 3) \stackrel{\text{set } = 0}{\Rightarrow} x = -3, 6 \end{aligned}$$

② Compute the sign of $f'(-3)$ and $f'(6)$

$$\begin{aligned} f''(x) &= e^x(x^2 - 3x - 18) + e^x(2x - 3) \\ &= e^x(x^2 - x - 21) \end{aligned}$$

$$f''(-3) = e^{-3}(9 + 3 - 21) = -9e^{-3} < 0 \quad \text{---}$$

$$f''(6) = e^6(36 - 6 - 21) > 0 \quad \text{++}$$

$\Rightarrow$ rel. max at $x = -3$ and rel min at $x = 6$